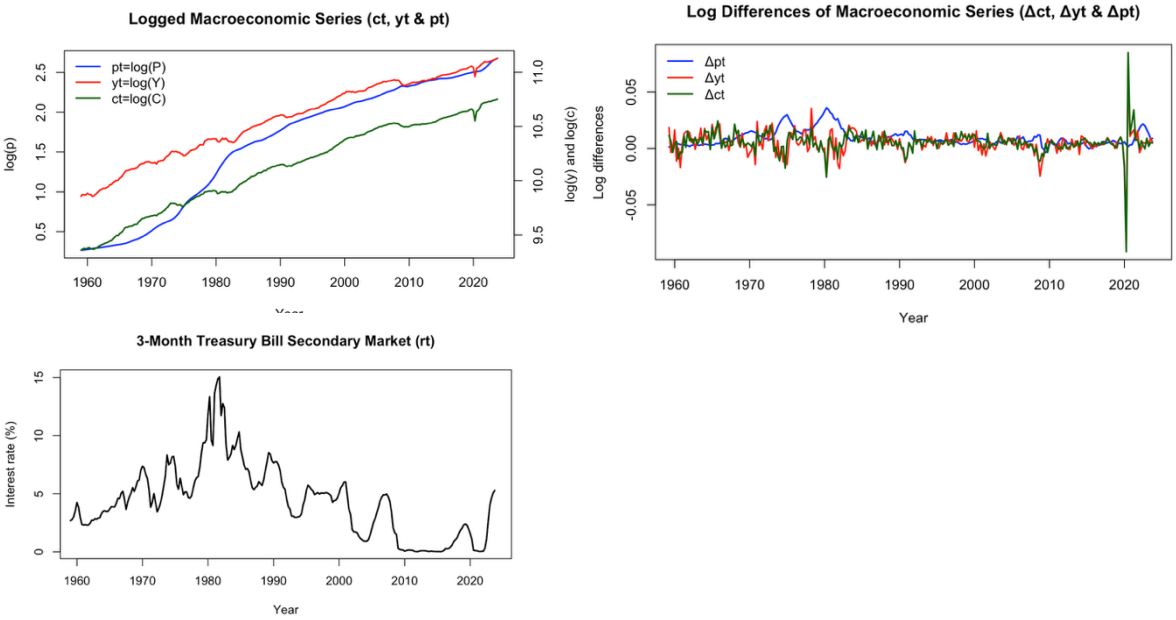
1a)



The plots of p_t, y_t and c_t in log levels all show persistent upward movement over time. This suggests that the processes generating these series have a trend in the process mean and so are not covariance stationary in levels. The upward movement is especially clear for y_t and c_t, which move closely together over the sample, while $p_t$also rises strongly, particularly through the 1970s and early 1980s. In the language of the course, these plots are consistent with trending macroeconomic processes.

The plots of Δp_t, Δy_t and Δc_t look much more stable than the level series. They fluctuate around fairly constant averages and do not show the same persistent upward movement, suggesting that differencing removes most of the trend. There are still some periods of larger volatility, especially around 2020, where Δc_tshows a very large temporary disturbance. Overall, this is consistent with the idea that the level processes may be difference stationary, or at least close to processes with stochastic trends, so that the first differences are much more stable than the levels.

The interest rate series $r_t$behaves differently from the other three series. It does not show the same steady upward trend in levels but instead displays long swings over time. Specifically, it rises into the late 1970s and early 1980s, declines over later decades, stays near zero for much of the 2010s, and then rises again after 2021. This suggests a persistent process with substantial autocorrelation and changes in its average level across different periods. So, the dominant feature of r_t is not steady growth, but prolonged movements over time.

1bi)

$$p_t = 0.270678 + 0.010125\, t + \varepsilon_t$$
$$y_t = 9.944853 + 0.004757\, t + \varepsilon_t$$
$$c_t = 9.429001 + 0.005305\, t + \varepsilon_t$$

ii)

$$\hat{\mu}_{\Delta p_t} = 0.009310$$
$$\hat{\mu}_{\Delta y_t} = 0.004919$$
$$\hat{\mu}_{\Delta c_t} = 0.005379$$

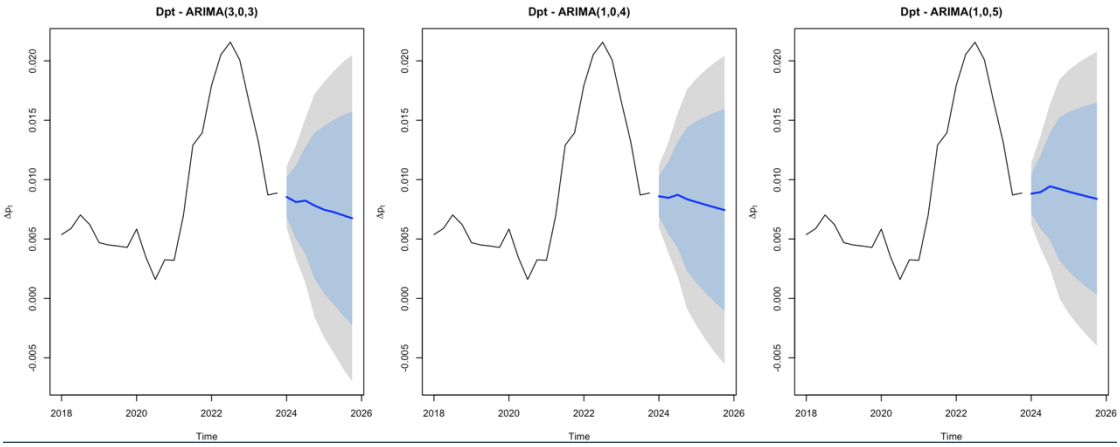
iii)

A rationale for these estimations is that they help characterise the trend behaviour of the processes. A persistent increase in the data suggests a trend in the underlying stochastic process, and modelling a deterministic trend in practice involves including time $t$as a regressor. Thus, regressing p_t, y_t and c_t on an intercept and time trend provides an estimate of the linear trend in the process mean, captured by $\hat{\delta}$.

By contrast, estimating the mean of Δp_t, Δy_t and Δc_t provides an estimate of the average quarter-to-quarter change in each series, captured by $\hat{\mu}$. If the log-level processes are trending and are reasonably close to difference-stationary behaviour, then the mean of the first differences should be similar to the slope of the linear trend in levels. So $\hat{\delta}$ and $\hat{\mu}$ are both trying to capture average growth, but they do so in slightly different ways. More specifically, $\hat{\delta}$ is through a fitted linear trend in the level series, and $\hat{\mu}$through the average change in the differenced series.

In this sample, the estimates are quite close. For y_t, the estimated time trend is 0.004757 while the mean of Δy_t is 0.004919. For c_t, the estimated time trend is 0.005305 while the mean of Δc_t is 0.005379. For p_t, the estimated time trend is 0.010125 while the mean of Δp_t is 0.009310. This similarity suggests that both approaches are capturing the same broad feature of the data, namely persistent average growth in the log-level series. The gap is slightly larger for p_t, which is consistent with the plot showing somewhat less smooth trend behaviour than for y_t and c_t. Overall, the results are consistent with the view that the level series are trending, while the first differences are much more stable. However, this comparison alone does not establish whether the processes contain unit roots; that requires formal unit root testing.

2a)



A broad set of ARIMA models was estimated for Δp_t and r_t, and the final three inflation models were selected using AIC and BIC, which balance fit and parsimony.

b) Forecasts of Δp_t are useful for policy because decisions depend not only on current conditions, but also on what is expected to happen in the future. In particular, medium-term inflation forecasts help policymakers assess the likely path of inflation and hence the appropriate policy stance. In the lecture language, a good forecasting methodology should not only match and extrapolate a pattern in the historical data, but also quantify the uncertainty associated with that prediction.

In this case, the three selected ARIMA models give very similar point forecasts. In 2024Q1 the forecasts are 0.00854, 0.00860 and 0.00881, and by 2025Q4 they are 0.00674, 0.00744 and 0.00838. So, the central outlook is broadly the same across models where inflation growth is expected to ease slightly from late-2023 levels and remain fairly stable through 2024 and 2025. The differences across the three point forecasts are small relative to the forecast intervals, suggesting limited model disagreement about the central path.

The main uncertainty comes from the unknown future, that is, future shocks that have not yet occurred. This is why the predictive intervals widen over the forecast horizon. For example, the 95% interval for ARIMA(3,0,3) widens from about $[0.00597, 0.01111]$ n 2024Q1 to about $[-0.00699, 0.02048]$ in 2025Q4. Similar widening occurs for the other two models. So even though the point forecasts remain near 0.007 to 0.008, the range of plausible outcomes becomes much wider by the end of 2025. This is consistent with the lecture point that point forecasts on their own are difficult to interpret and that uncertainty should be quantified through forecast error variance and predictive intervals.

There is also specification uncertainty where there are many plausible ARIMA models for the same process, and in practice there is no such thing as a true model. Further, there is estimation uncertainty such that even after choosing a specification, the coefficients are estimated from a finite sample and would differ across samples. Using three ARIMA models rather than one partly addresses specification uncertainty, while the predictive intervals capture the remaining uncertainty around the central forecast. Finally, these forecasts uses the assumption that the historical pattern remains valid in the future. If the economy moves into a different regime or is hit by a large, unexpected shock, realised inflation may differ from the forecasts.

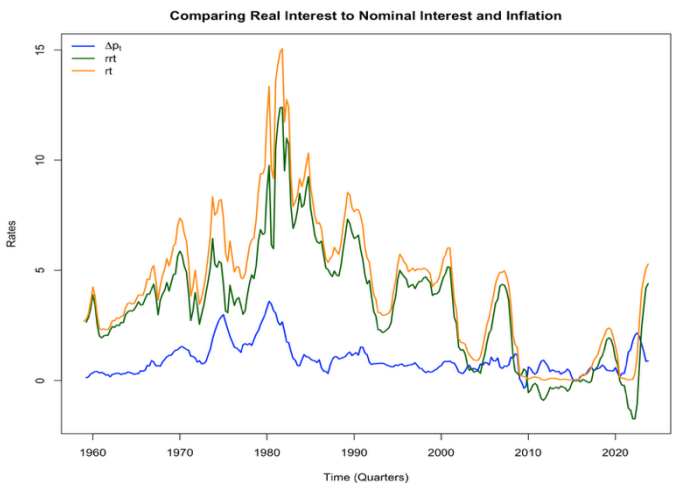
3) To evaluate the relative forecast performance of the three inflation models, the realised values for 2024Q1 to 2025Q3 are treated as a hold-out sample and compared with the forecasts from Question 2. Forecast performance is assessed using MSFE and MAFE. Lower values are preferred because they mean the forecasts are on average, closer to the realised outcomes. As for MSFE, larger forecast errors are penalised more heavily because they are squared.

Using these measures, ARIMA(3,0,3) performs best, with MSFE = 5.98×10^{-7} and MAFE = 0.000669. ARIMA(1,0,4) ranks second, with MSFE = 1.50×10^{-6} and MAFE = 0.001096, while ARIMA(1,0,5) performs worst, with MSFE = 3.82×10^{-6} and MAFE = 0.001806. Since both measures give the same ranking, the evidence is fairly clear that ARIMA(3,0,3) provides the best out-of-sample forecast performance over this period.

The figure supports the same conclusion where all three models capture the broad pattern in inflation reasonably well and the realised values remain inside the predictive intervals throughout the hold-out. However, the differences in forecast accuracy still matter. ARIMA(3,0,3) tracks the realised series more closely over most of 2024Q1 to 2025Q3, whereas ARIMA(1,0,4) and especially ARIMA(1,0,5) tend to sit above the realised values for more of the holdout period. In other words, the latter two models tend to over-predict inflation growth by more. This is why their MSFE and MAFE are larger.

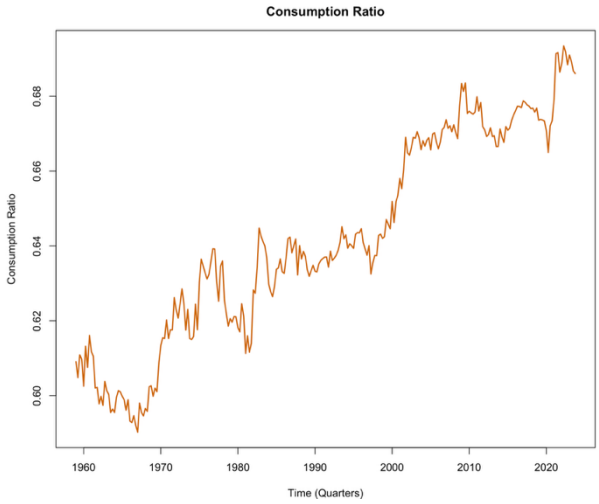
Overall, the three models produce similar central forecasts, and all retain reasonable predictive intervals, but ARIMA(3,0,3) gives the smallest forecast errors and the closest match to the realised inflation data. For this application, it is therefore the preferred forecasting model of the three considered.

4a)



The plot suggests that the real interest rate proxy rr_t is somewhat more stable than the nominal interest rate r_t, but less stable than inflation Δp_t. The nominal interest rate shows the largest long swings, particularly in the late 1970s and early 1980s, while the real interest rate is less extreme because inflation has been removed. However, rr_t still displays substantial variation over time, including prolonged swings and some negative values, so it is not obviously stable in any strong sense. Relative to the other two series, inflation appears to fluctuate within the narrowest range for most of the sample. Hence, the evidence from the plot is that rr_t is less volatile than r_t, but not clearly more stable than Δp_t.

b)



The dominant feature of $cy_t = \frac{C_t}{Y_t}$ is its persistent upward movement over the sample. This persistent increase in the realised data suggests a trend in the underlying stochastic process, so the process mean does not appear constant over time. The series does not fluctuate around a clearly fixed level. Instead, the consumption ratio rises from about 0.60 in the early part of the sample to around 0.69 by the end, with only moderate short-run fluctuations around this longer-run movement. Persistent increases of this kind are exactly the sort of pattern Rodney treats as evidence of a trend in the process mean.

From an economic perspective, the increase in cy_t means that consumption has grown faster than output over long parts of the sample, so a larger share of GDP has been allocated to consumption. Since $cy_t = \frac{C_t}{Y_t}$, a rise in the ratio means that C_t is increasing relative to Y_t. Equivalently, from the macro identity $Y = C + I + G + (X - M)$, a higher consumption share implies that the share of output accounted for by the other components has, on average, fallen relative to consumption. So, the dominant feature of the plot is not high volatility, but a slow-moving upward trend or structural shift in the consumption share.

c) A broad set of ARIMA models was estimated for rr_t and compared using AIC and BIC. Although ARIMA(3,0,2) achieved the lowest BIC, it was not adequate because the Ljung–Box test gave a p-value of 0.0103, indicating remaining residual autocorrelation. Another candidate model was also inadequate, with a p-value of 0.0004894. By contrast, ARIMA(8,0,1) produced a Ljung–Box p-value of 0.7301, so there is no evidence of residual autocorrelation in its residuals. Hence, ARIMA(8,0,1) is the best adequate model for rr_t.

This model is also consistent with the main features of the plot in Question 4(a). The fact that $d = 0$ means the inferred process for rr_t did not require differencing, which fits the graph where the real interest rate appears more stable than the strongly trending series considered earlier. At the same time, the relatively high autoregressive order indicates substantial persistence, so rr_t still displays long swings over time rather than fluctuating independently from one quarter to the next. The moving-average term captures short-run dependence in shocks. Overall, ARIMA(8,0,1) captures the fact that rr_t is not dominated by a strong trend but is still persistent and serially dependent.

d) A broad set of ARIMA models was estimated for cy_t and compared using AIC and BIC. The lowest-BIC candidate was **ARIMA(1,0,1)**. If its residual diagnostics indicate no remaining autocorrelation, then it is the preferred model because it is both **adequate** and **parsimonious**. This follows the general model-selection strategy of first identifying an adequate set of models and then choosing the best model within that set using information criteria.

The model also matches the dominant feature of the plot in Question 4(b). The consumption ratio shows a persistent upward movement, so the realised data do not fluctuate around a clearly fixed level. The fact that the selected model has $d = 0$ means the inferred process did not require differencing, even though the series is clearly persistent. The AR and MA terms capture this persistence and short-run dependence, so the model is consistent with a process that is dominated by slow-moving behaviour rather than large short-run volatility.

e) The model for rr_t may be useful for policy because the real interest rate is closely related to the effective stance of monetary policy. A forecasting model for rr_t can help policymakers assess whether real borrowing conditions are expected to remain tight or loose, which matters for consumption, investment, and aggregate demand. In this application, the selected model for rr_t captures persistence and long swings in the series, so it is useful for summarising how real financing conditions evolve over time rather than treating each quarterly movement as independent. More generally, forecasts are useful because decisions depend not only on current conditions, but also on what is expected to happen in the future.

The model for cy_t may be useful for policy because the consumption ratio provides information about the share of output allocated to consumption. Since the plot suggests persistent slow-moving behaviour in cy_t, the model can help policymakers assess whether the composition of GDP is shifting over time, which is relevant for understanding household demand and broader macroeconomic balance. In particular, if the consumption share is expected to remain high or continue rising, that may have implications for the relative roles of saving, investment, and external balance. So, while the rr_t model is useful for monitoring real financial conditions, the cy_t model is useful for monitoring persistent changes in the structure of demand.

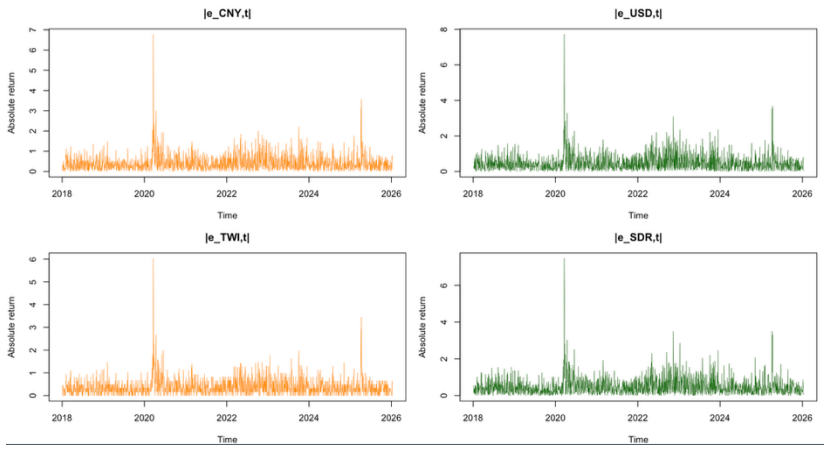
5a) $$\sigma^2_{CNY} = 0.3369 \;\; \sigma^2_{USD} = 0.4381 \; \sigma^2_{TWI} = 0.2675 \; \sigma^2_{SDR} = 0.4492$$

These **sample** variances estimate the unconditional variances of the return series and therefore summarise their average volatility over the sample. Since variance is a second-moment measure of dispersion, a larger value implies greater average variability in returns. The estimates show that SDR has the highest unconditional variance, followed closely by USD, while TWI has the lowest and CNY lies in between. Hence, the ranking of volatility is

$$SDR > USD > CNY > TWI.$$

This suggests that the AUD/SDR and AUD/USD return processes are the most volatile on average, while AUD/TWI is the least volatile. However, these are only unconditional summaries and do not show whether volatility is constant over time.

b)



The plots of | $e_{j,t}$ | show clear periods of low volatility followed by periods of much higher volatility, especially around 2020 and again in later parts of the sample. Large absolute returns also tend to be followed by further large absolute returns, while small absolute returns tend to be followed by further small absolute returns. This is evidence of volatility clustering. As a result, the return processes do not appear consistent with an i.i.d. process with constant variance. Instead, the plots suggest that the variance changes over time, so there is evidence of heteroskedasticity. The implication for modelling $e_{j,t}$ is that a constant-variance model would be inadequate, and a model with time-varying conditional variance, such as ARCH or GARCH, is more appropriate.

6) A broad set of ARMA(p, q)-GARCH(p_σ, q_σ) models was estimated for each return series $e_{j,t}$. Model selection was based on two ideas: fit versus parsimony by using AIC and BIC, and adequacy of the residuals. Lower AIC and BIC are preferred, but only within the set of models that appear adequate. Adequacy was assessed using Ljung-Box tests on standardized residuals and squared standardized residuals, since we are not to leave important autocorrelation in either the conditional mean or the conditional variance

For CNY, the selected model is ARMA(2,2)-GARCH(1,1), with AIC $= 1.555$, BIC $= 1.577$, Ljung-Box p-value on standardized residuals $= 0.609$, and Ljung-Box p-value on squared standardized residuals $= 0.269$. For USD, the selected model is ARMA(0,0)-GARCH(1,2), with AIC $= 1.839$, BIC $= 1.853$, and corresponding Ljung-Box p-values of $0.905$and 0.192. For TWI, the selected model is also ARMA(0,0)-GARCH(1,2), with AIC $= 1.315$, BIC $= 1.329$, and Ljung-Box p-values of $0.601$and 0.288. For SDR, the selected model is ARMA(2,2)-GARCH(1,2), with AIC $= 1.824$, BIC $= 1.849$, and Ljung-Box p-values of $0.676$and 0.060. Since all of these p-values are above 0.05, there is no strong evidence of remaining autocorrelation in the standardized residuals or in the squared standardized residuals, so these models are adequate by the main diagnostics used here.

These results are also consistent with the return plots from Question 5. The absolute returns showed clear volatility clustering, so a constant-variance model would be inadequate. The selected GARCH models allow the conditional variance to change over time, which is exactly the type of heteroskedasticity seen in financial data. In particular, the estimated volatility plots should mirror the bursts of volatility seen in $|\ e_{j,t}\ |$, with sharp spikes in stressed periods and quieter intervals elsewhere. The mean equations also differ across series in an intuitive way: CNY and SDR require richer ARMA dynamics in the conditional mean, whereas USD and TWI are adequately described with no ARMA terms in the mean and time-varying volatility only.

There is also evidence supporting the presence of GARCH-type effects. The volatility clustering observed in Question 5 is already strong visual evidence against homoskedasticity. In addition, the ARCH-LM statistics are relatively small in p-value for several series, especially CNY (0.049), USD (0.029) and SDR (0.0089), while TWI (0.068) is weaker. This suggests that time-varying conditional variance is an important feature of these return processes and helps justify the use of GARCH-type specifications rather than constant-variance ARMA models. At the same time, the SDR model is the least clean of the four, since its squared-residual Ljung-Box p-value of $0.060$is only just above 0.05, so its adequacy should be interpreted a little more cautiously than the others.

7) $\sigma^2_{CNY} = 0.3369 \;\; \sigma^2_{USD} = 0.4381 \; \sigma^2_{TWI} = 0.2675 \; \sigma^2_{SDR} = 0.4492$

Using the selected ARMA(p, q)-GARCH(p_σ, q_σ) models from Question 6, the corresponding model-implied estimates of the unconditional variances were:

$$\hat{\sigma}^2_{CNY,\text{model}} = 0.3106, \hat{\sigma}^2_{USD,\text{model}} = 0.4337, \hat{\sigma}^2_{TWI,\text{model}} = 0.2418, \hat{\sigma}^2_{SDR,\text{model}} = 0.4479$$

Comparing these estimates with those from Question 5, the model-based unconditional variances are close to the sample variances for all four series. For CNY, the model estimate is slightly lower than the sample estimate, 0.3106 compared with 0.3369. For USD, the two are almost identical, 0.4337 versus 0.4381. For TWI, the model estimate 0.2418 is below the sample estimate 0.2675, giving the largest gap of the four series, although the two values are still reasonably close. For SDR, the agreement is again very strong, with 0.4479 from the model compared with 0.4492 from the sample. Overall, the model-based estimates preserve the same broad ranking of volatility found in Question 5, namely that SDR and USD are the most volatile, CNY is intermediate, and TWI is the least volatile.

In every case, the model-implied unconditional variance is slightly smaller than the raw sample variance. A sensible interpretation is that the fitted GARCH models smooth the volatility process and estimate the long-run variance implied by the conditional variance dynamics, whereas the sample variance is influenced more directly by particular large, realised movements in the data. Even so, the fact that the two sets of estimates are close suggests that the selected ARMA-GARCH models provide a reasonable description of the long-run volatility of each exchange-rate return process.

It is possible to compute $\hat{\sigma}^2_{j,\text{model}}$ for all four series because the estimated persistence of each GARCH model is below one specifically where it was 0.924 for CNY, 0.962 for USD, 0.905 for TWI and 0.966 for SDR. As they were less than one, the models imply a finite unconditional variance, so there are no cases here in which the unconditional variance cannot be computed. At the same time, the persistence values are all quite high, especially for USD and SDR, which indicates that volatility shocks are very persistent and decay only gradually over time. This is consistent with the volatility clustering observed earlier in the absolute return plots.

8) The estimated probabilities on 13/01/2026 and 14/01/2026 where the returns are less than 0.01% using the ARMA-GARCH models are:

$$\Pr\left(e_{CNY,t} < 0.01\right) = 0.5207 \text{ on 13/01/2026, and } 0.5202 \text{ on 14/01/2026}$$
$$\Pr\left(e_{USD,t} < 0.01\right) = 0.5167 \text{ on 13/01/2026, and } 0.5171 \text{ on 14/01/2026}$$
$$\Pr\left(e_{TWI,t} < 0.01\right) = 0.5134 \text{ on 13/01/2026, and } 0.5131 \text{ on 14/01/2026}$$
$$\Pr\left(e_{SDR,t} < 0.01\right) = 0.5189 \text{ on 13/01/2026, and } 0.5159 \text{ on 14/01/2026}$$

The probabilities are all slightly above 0.5. This happens because the forecast conditional means are negative for all four series, while the threshold 0.01% is close to zero. So, on both dates, there is a slightly greater than 50% chance that the return falls below the threshold.

Comparing across currencies, TWI has the lowest probability of a return below 0.01%on both days. On 13/01/2026, the ranking from lowest to highest downside probability is:

$$TWI < USD < SDR < CNY$$

and on 14/01/2026 the ranking is:

$$TWI < SDR < USD < CNY$$

So, TWI is the preferred currency from a downside-risk perspective, while CNY is the least attractive, since it has the highest probability of falling below the threshold on both days. The main reason is that TWI combines the least negative forecast mean with the lowest forecast conditional volatility, whereas CNY has a more negative forecast mean and relatively high volatility.

In risk-management terms, this information can be used to rank the currencies according to their short-run downside risk. A more risk-averse investor would prefer the currency with the smallest value of $\Pr\left(e_{j,t} < 0.01\right)$, since that implies the lowest probability of underperforming the threshold. Hence, based on these results, I would choose AUD/TWI as the safest of the four options for these two days. More generally, this exercise shows how conditional mean and conditional variance forecasts can be combined to form predictive probabilities, which are directly useful for risk management and investment decisions.